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MA3521 — In-Class Worksheet Thursday/Monday
Do not row reduce or solve.

Name:

1. $\mathbf{y}' = \boldsymbol{A}\mathbf{y} + e^{-2t}\begin{pmatrix}1\\2\\3\end{pmatrix}$ $\quad \boldsymbol{A} = \begin{pmatrix}3 & -4 & 4\\0 & -1 & 3\\0 & 0 & 2\end{pmatrix}$ $\quad \mathbf{y}(0) = \mathbf{y}_0 = \begin{pmatrix}1\\0\\2\end{pmatrix}$

   Eigenpairs -1, $\begin{pmatrix}1\\1\\0\end{pmatrix}$ $\quad 2$, $\begin{pmatrix}0\\1\\1\end{pmatrix}$ $\quad 3$, $\begin{pmatrix}1\\0\\0\end{pmatrix}$

   (a) $\mathbf{y}_c(t) =$

   (b) A suitable UC form is

   $\mathbf{y}_p(t) =$

   (c) To determine the UC coefficients you would row reduce

   $\boldsymbol{M}_{\text{UC}} =$

   (d) To satisfy the IC you would row reduce

   $\boldsymbol{M}_{\text{IC}} =$

Not on exam $\mathbf{y}(t) =$

2. $\mathbf{y}' = \boldsymbol{A}\mathbf{y} + \cos(3t)\begin{pmatrix}1\\2\end{pmatrix} \quad \boldsymbol{A} = \begin{pmatrix}2 & -6\\3 & -4\end{pmatrix} \quad \mathbf{y}(0) = \mathbf{y}_0 = \begin{pmatrix}1\\0\end{pmatrix}$

Eigenpairs $-1+3i$, $\begin{pmatrix}2\\1-i\end{pmatrix}$ $\quad -1-3i$, $\begin{pmatrix}2\\1+i\end{pmatrix}$

(a) $\mathbf{y}_c(t) =$

(b) A suitable UC form is

$\mathbf{y}_p(t) =$

(c) To determine the UC coefficients you would row reduce

$\boldsymbol{M}_{\text{UC}} =$

(d) To satisfy the IC you would row reduce

$\boldsymbol{M}_{\text{IC}} =$

Not on exam $\mathbf{y}(t) =$